

score__forecasts

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Load in truth data and forecasts

We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

Plot Forecasts

We plot forecasts for three locations (Us National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.

```
locs <- full_hosp_truth %>%
  filter(target_end_date <= as.Date("2021-07-05")) %>%
  group_by(location) %>%
  summarize(cum_value=sum(value)) %>%
  ungroup() %>%
  arrange(desc(cum_value)) %>%
  filter(row_number() %in% c(1, 2, 53)) %>%
  pull(location)

fc_dates <- c(as.Date("2020-12-07") + weeks(4*(0:7)))

fc_df <- forecasts_df %>% rbind(forecasts_baseline)
for (i in 1:3) {
  fdat <- fc_df %>%
    filter(location == locs[i], forecast_date %in% fc_dates)

  p <- plot_forecasts(fdat,
#       truth_data = filter(full_hosp_truth, target_end_date <= as.Date("2021-05-03")),
    target_variable = "inc hosp",
    intervals = c(.5, .95),
    truth_source = "HealthData",
    use_median_as_point = TRUE,
    facet = model ~.,
    facet_nrow = 4,
    #facet_scales = "free_y",
    fill_by_model = TRUE,
    plot=FALSE)
```

```

pt <- p +
  scale_x_date(name=NULL, limits=c(as.Date("2020-10-01"), as.Date("2021-07-05")),
    date_breaks = "2 months", date_labels = "%b") +
  #scale_y_continuous(limits=c(0, max(filter(full_hosp_truth, location==locs[i])$value) * 1.25)) +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2),
    legend.position = "none")

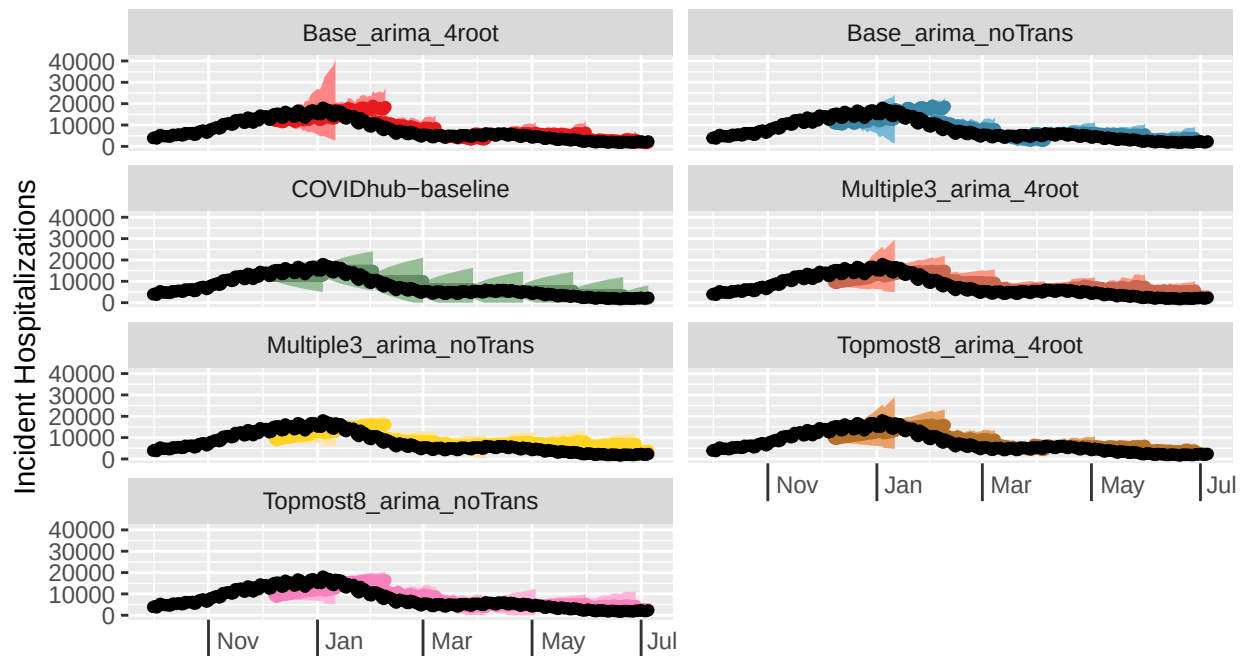
print(pt)
}

```

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): United States

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-04-01, 2021-05-01, 2021-06-01, 2021-07-01

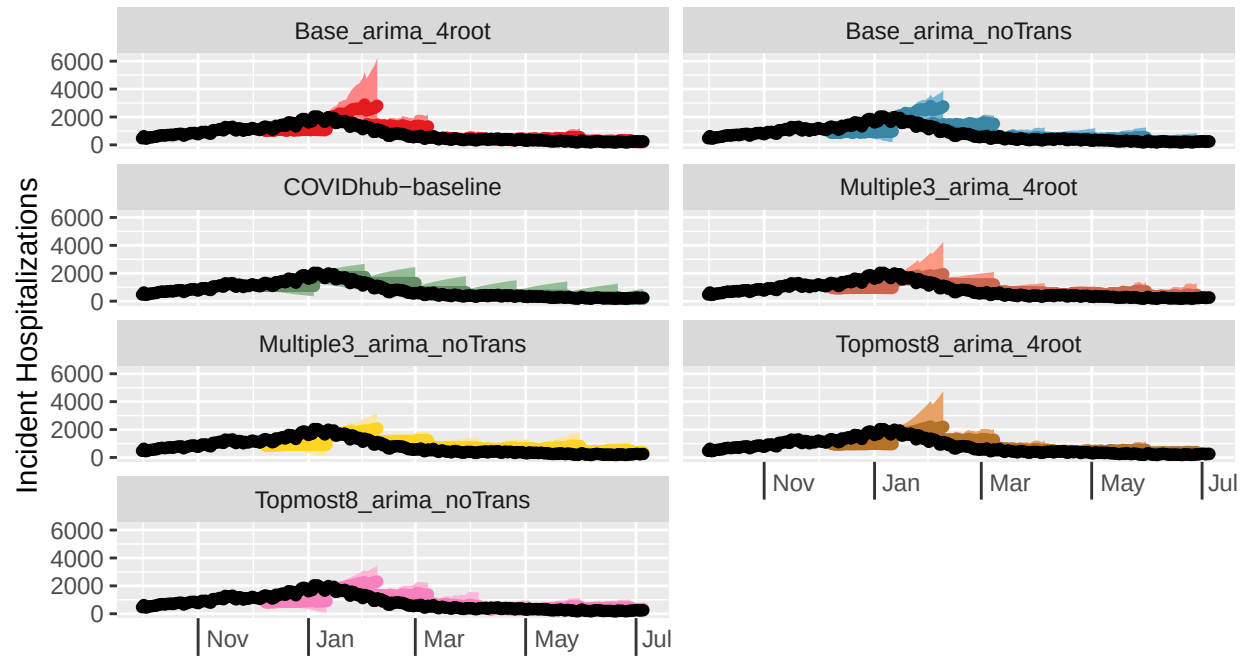


4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans, COVIDhub-baseline (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Texas

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021

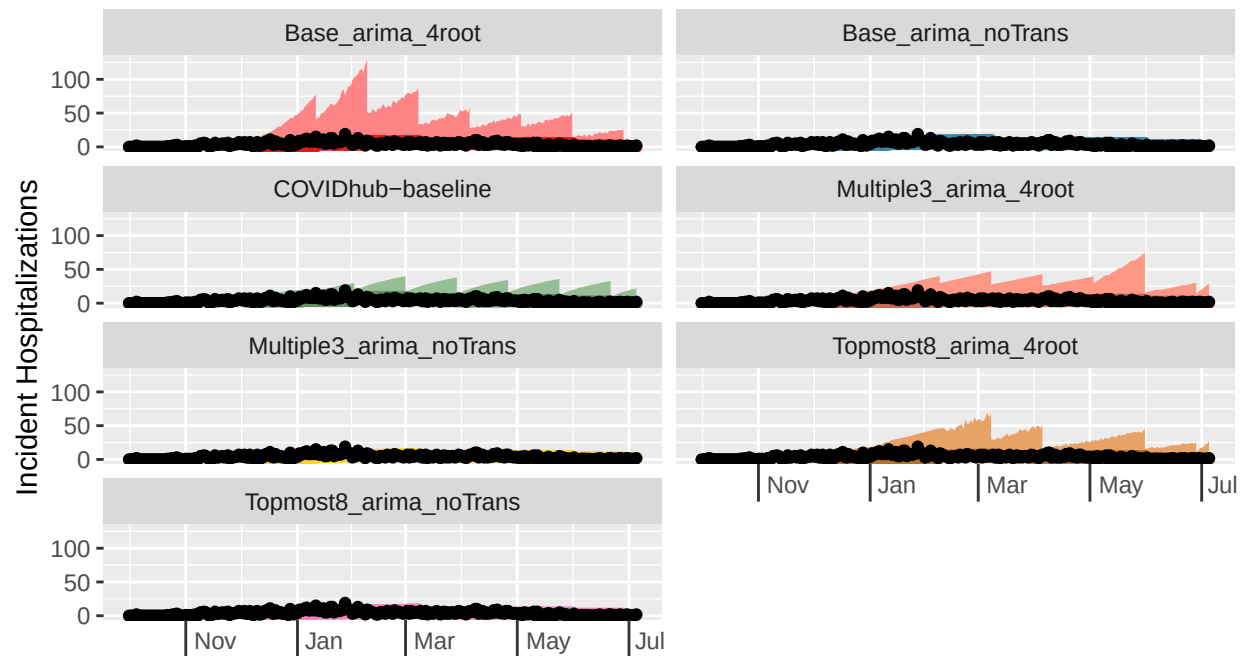


4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans, COVIDhub-baseline (forecasts)

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): Vermont

Selected forecast date(s): 2020-12-07, 2021-01-04, 2021-02-01, 2021-03-01, 2021-



4root, Multiple3_arma_noTrans, Topmost8_arma_4root, Topmost8_arma_noTrans, COVIDhub-baseline (forecasts)

The plots don't show much difference between the 6 models (although this is partially due to the fact that the yscals are free and are not consistent even for the same location). However, of note is the fact that the models that use a forth root transformation tend to have wider prediction intervals than the models without transformations. All of the models reasonably follow the trend of the truth lines for every location shown, although it seems like lower count locations may have slightly greater accuracy. The baseline seems to have intervals of fairly consistent widths while the models tend to vary in their interval widths, tending to show wider intervals during times of higher incidents or change points and narrow lower intervals otherwise. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

```
#scores_df <- score_forecasts(forecasts=forecasts_df, return_format="wide", truth=full_hosp_truth, use_
#save(scores_df, file="../data/loadedfc_df.RData")
# score_baseline <- score_forecasts(forecasts=forecasts_baseline, return_format="wide", truth=full_hosp
#save(forecasts_baseline, score_baseline, file="../data/baseline_fc_scores.RData")
```

Overall metrics

```
# US National
overall_metrics_US <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location == "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    )
  ) %>%
  group_by(model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    cov_50 = mean(coverage_50),
    cov_95 = mean(coverage_95)
  )
```

'summarise()' has grouped output by 'model'. You can override using the '.groups' argument.

```
overall_metrics_US <- overall_metrics_US %>%
  mutate(
    rwis = case_when(
      horizon_wk == 1 ~ wis / pull(overall_metrics_US[9,3]),
      horizon_wk == 2 ~ wis / pull(overall_metrics_US[10,3]),
      horizon_wk == 3 ~ wis / pull(overall_metrics_US[11,3]),
      horizon_wk == 4 ~ wis / pull(overall_metrics_US[12,3])
    ),
    rmae = case_when(
      horizon_wk == 1 ~ mae / pull(overall_metrics_US[9,4]),
      horizon_wk == 2 ~ mae / pull(overall_metrics_US[10,4]),
      horizon_wk == 3 ~ mae / pull(overall_metrics_US[11,4]),
      horizon_wk == 4 ~ mae / pull(overall_metrics_US[12,4])
    ),
    rcov_50 = case_when(
      horizon_wk == 1 ~ (cov_50-0.5) / abs(pull(overall_metrics_US[9,5])-0.5),
      horizon_wk == 2 ~ (cov_50-0.5) / abs(pull(overall_metrics_US[10,5])-0.5),
      horizon_wk == 3 ~ (cov_50-0.5) / abs(pull(overall_metrics_US[11,5])-0.5),
      horizon_wk == 4 ~ (cov_50-0.5) / abs(pull(overall_metrics_US[12,5])-0.5),
    ),
    rcov_95 = case_when(
      horizon_wk == 1 ~ (cov_95-0.95) / abs(pull(overall_metrics_US[9,6])-0.95),
      horizon_wk == 2 ~ (cov_95-0.95) / abs(pull(overall_metrics_US[10,6])-0.95),
      horizon_wk == 3 ~ (cov_95-0.95) / abs(pull(overall_metrics_US[11,6])-0.95),
      horizon_wk == 4 ~ (cov_95-0.95) / abs(pull(overall_metrics_US[12,6])-0.95),
    )
  ) %>%
```

```
mutate(across(where(is.numeric), round, digits=3))
knitr::kable(overall_metrics_US)
```

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arma_4root	1	542.076	836.752	0.117	0.800	0.897	1.081	-1.278	-9.000
Base_arma_4root	2	860.165	1323.950	0.393	0.841	0.822	0.856	-0.373	-2.172
Base_arma_4root	3	1156.882	1765.098	0.337	0.827	0.899	0.914	-0.640	-2.469
Base_arma_4root	4	1567.629	2346.378	0.259	0.772	1.001	0.983	-1.123	-3.550
Base_arma_noTrans	1	716.236	1063.424	0.083	0.783	1.186	1.374	-1.389	-
									10.000
Base_arma_noTrans	2	958.092	1359.289	0.476	0.828	0.915	0.879	-0.084	-2.448
Base_arma_noTrans	3	1200.986	1718.652	0.490	0.847	0.933	0.890	-0.040	-2.061
Base_arma_noTrans	4	1551.446	2292.973	0.370	0.857	0.990	0.961	-0.605	-1.857
COVIDhub-baseline	1	604.156	773.900	0.800	0.967	1.000	1.000	1.000	1.000
COVIDhub-baseline	2	1046.662	1546.717	0.786	1.000	1.000	1.000	1.000	1.000
COVIDhub-baseline	3	1287.123	1930.434	0.755	1.000	1.000	1.000	1.000	1.000
COVIDhub-baseline	4	1566.581	2385.804	0.714	1.000	1.000	1.000	1.000	1.000
Multiple3_arma_4root	1	1647.772	2105.831	0.050	0.250	2.727	2.721	-1.500	-
									42.000
Multiple3_arma_4root	2	1396.428	2135.817	0.145	0.628	1.334	1.381	-1.241	-6.448
Multiple3_arma_4root	3	1416.465	2197.313	0.224	0.709	1.100	1.138	-1.080	-4.816
Multiple3_arma_4root	4	1763.813	2694.806	0.233	0.582	1.126	1.130	-1.247	-7.360
Multiple3_arma_noTrans	1	1722.915	2232.146	0.083	0.317	2.852	2.884	-1.389	-
									38.000
Multiple3_arma_noTrans	2	1430.444	2300.854	0.097	0.766	1.367	1.488	-1.410	-3.690
Multiple3_arma_noTrans	3	1482.037	2403.128	0.219	0.852	1.151	1.245	-1.100	-1.959
Multiple3_arma_noTrans	4	1906.551	3012.165	0.201	0.762	1.217	1.263	-1.395	-3.762
Topmost8_arma_4root	1	1116.034	1512.255	0.067	0.333	1.847	1.954	-1.444	-
									37.000
Topmost8_arma_4root	2	1044.122	1653.197	0.234	0.759	0.998	1.069	-0.928	-3.828
Topmost8_arma_4root	3	1212.164	1919.171	0.250	0.755	0.942	0.994	-0.980	-3.898
Topmost8_arma_4root	4	1470.919	2299.869	0.254	0.683	0.939	0.964	-1.148	-5.349
Topmost8_arma_noTrans	1	1056.776	1442.944	0.200	0.700	1.749	1.865	-1.000	-
									15.000
Topmost8_arma_noTrans	2	1129.085	1778.506	0.324	0.814	1.079	1.150	-0.614	-2.724
Topmost8_arma_noTrans	3	1298.125	2080.367	0.321	0.847	1.009	1.078	-0.700	-2.061
Topmost8_arma_noTrans	4	1588.545	2432.356	0.381	0.831	1.014	1.020	-0.556	-2.386

```
# Aggregated for all states and territories
overall_metrics_states <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location != "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    )
  ) %>%
```

```
group_by(model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    cov_50 = mean(coverage_50),
    cov_95 = mean(coverage_95)
  )
```

'summarise()' has grouped output by 'model'. You can override using the '.groups' argument.

```
overall_metrics_states <- overall_metrics_states %>%
  mutate(
    rwis = case_when(
      horizon_wk == 1 ~ wis / pull(overall_metrics_states[9,3]),
      horizon_wk == 2 ~ wis / pull(overall_metrics_states[10,3]),
      horizon_wk == 3 ~ wis / pull(overall_metrics_states[11,3]),
      horizon_wk == 4 ~ wis / pull(overall_metrics_states[12,3])
    ),
    rmae = case_when(
      horizon_wk == 1 ~ mae / pull(overall_metrics_states[9,4]),
      horizon_wk == 2 ~ mae / pull(overall_metrics_states[10,4]),
      horizon_wk == 3 ~ mae / pull(overall_metrics_states[11,4]),
      horizon_wk == 4 ~ mae / pull(overall_metrics_states[12,4])
    ),
    rcov_50 = case_when(
      horizon_wk == 1 ~ (cov_50-0.5) / abs(pull(overall_metrics_states[9,5])-0.5),
      horizon_wk == 2 ~ (cov_50-0.5) / abs(pull(overall_metrics_states[10,5])-0.5),
      horizon_wk == 3 ~ (cov_50-0.5) / abs(pull(overall_metrics_states[11,5])-0.5),
      horizon_wk == 4 ~ (cov_50-0.5) / abs(pull(overall_metrics_states[12,5])-0.5),
    ),
    rcov_95 = case_when(
      horizon_wk == 1 ~ (cov_95-0.95) / abs(pull(overall_metrics_states[9,6])-0.95),
      horizon_wk == 2 ~ (cov_95-0.95) / abs(pull(overall_metrics_states[10,6])-0.95),
      horizon_wk == 3 ~ (cov_95-0.95) / abs(pull(overall_metrics_states[11,6])-0.95),
      horizon_wk == 4 ~ (cov_95-0.95) / abs(pull(overall_metrics_states[12,6])-0.95),
    )
  ) %>%
  mutate(across(where(is.numeric), round, digits=3))
knitr::kable(overall_metrics_states)
```

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
Base_arma_4root	1	17.822	25.568	0.504	0.892	0.958	1.102	0.010	-2.275
Base_arma_4root	2	24.224	37.394	0.486	0.928	0.799	0.957	-0.035	-0.506
Base_arma_4root	3	31.057	47.771	0.480	0.923	0.853	0.997	-0.051	-0.622
Base_arma_4root	4	39.583	60.273	0.462	0.906	0.920	1.030	-0.096	-1.049
Base_arma_noTrans	1	20.333	28.783	0.523	0.903	1.093	1.240	0.067	-1.850
Base_arma_noTrans	2	26.076	39.368	0.577	0.928	0.860	1.008	0.195	-0.512
Base_arma_noTrans	3	32.443	48.423	0.590	0.918	0.891	1.010	0.227	-0.741
Base_arma_noTrans	4	40.149	59.564	0.581	0.906	0.933	1.018	0.207	-1.059
COVIDhub-baseline	1	18.603	23.209	0.837	0.976	1.000	1.000	1.000	1.000
COVIDhub-baseline	2	30.328	39.058	0.896	0.994	1.000	1.000	1.000	1.000
COVIDhub-baseline	3	36.412	47.923	0.896	0.993	1.000	1.000	1.000	1.000

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
COVIDhub-baseline	4	43.018	58.502	0.891	0.992	1.000	1.000	1.000	1.000
Multiple3_arima_4root	1	35.491	47.142	0.317	0.720	1.908	2.031	-0.542	-8.975
Multiple3_arima_4root	2	34.463	49.263	0.360	0.853	1.136	1.261	-0.353	-2.227
Multiple3_arima_4root	3	45.006	52.951	0.392	0.884	1.236	1.105	-0.272	-1.533
Multiple3_arima_4root	4	90.817	63.989	0.380	0.873	2.111	1.094	-0.306	-1.825
Multiple3_arima_noTrans	1	35.154	47.842	0.315	0.762	1.890	2.061	-0.548	-7.350
Multiple3_arima_noTrans	2	33.424	52.824	0.372	0.929	1.102	1.352	-0.323	-0.485
Multiple3_arima_noTrans	3	36.466	58.694	0.408	0.940	1.002	1.225	-0.231	-0.237
Multiple3_arima_noTrans	4	44.741	72.387	0.394	0.931	1.040	1.237	-0.272	-0.445
Topmost8_arima_4root	1	26.963	37.196	0.382	0.788	1.449	1.603	-0.349	-6.300
Topmost8_arima_4root	2	26.540	41.036	0.414	0.902	0.875	1.051	-0.216	-1.100
Topmost8_arima_4root	3	30.690	47.891	0.425	0.914	0.843	0.999	-0.191	-0.830
Topmost8_arima_4root	4	36.822	57.179	0.425	0.908	0.856	0.977	-0.191	-1.005
Topmost8_arima_noTrans	1	25.963	35.412	0.464	0.862	1.396	1.526	-0.107	-3.437
Topmost8_arima_noTrans	2	27.778	41.741	0.531	0.927	0.916	1.069	0.079	-0.527
Topmost8_arima_noTrans	3	32.357	49.562	0.532	0.928	0.889	1.034	0.082	-0.521
Topmost8_arima_noTrans	4	37.998	58.305	0.548	0.919	0.883	0.997	0.124	-0.741

When splitting the results so that US National is separate, we find that for US National the two models that use the base set hierarchy tended to perform the best, with improvements seen over a naive baseline in all metrics except for 95%PI Coverage. These two models are followed by the Topmost8_arima_noTrans model, which showed more comparable performance to the baseline. The Base_arima_noTrans model performed the best in terms of PI Coverage, followed by the Base_arima_4root and Topmost8_arima_noTrans models (with the latter being more consistent with coverage rates for all horizons). In terms of WIS and MAE, the Base_arima_4root model performed the best, followed by the Base_arima_noTrans model. Overall, for the US National level, it seems that there is a pattern that for coverage rates, the models without transformations perform better (and these tend to have smaller prediction intervals) while for WIS and MAE the fourth root transform models perform better; these patterns hold at all horizons. Further, there is a general order to model accuracy in terms of hierarchical structure used: the Base set performs the best, followed by the Topmost8 set, and finally the Multiple3 set.

For the state level, we see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models. (We also see shrunk WIS and MAE values, although this is likely partially due to the fact that the states have smaller scale values than the National level.) Interestingly, we see fewer improvements to MAE over the baseline but more substantial improvements in terms of coverage rates. It's a little harder to definitively say which models have the best coverage rates, as the results differ between the 50% and 95% intervals and the models are quite similar in coverage rates, especially for the 95% intervals. However, it seems that the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model.

Metrics by wave

```
# US National
wave_metrics_US <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location == "US") %>%
```



```

mutate(
  horizon_wk = case_when(
    horizon %in% c(1:7) ~ 1,
    horizon %in% c(8:14) ~ 2,
    horizon %in% c(15:21) ~ 3,
    horizon %in% c(22:28) ~ 4
  ),
  wave = case_when(
    forecast_date <= as.Date("2021-03-17") ~ "winter21",
    forecast_date >= as.Date("2021-03-17") ~ "alpha"
  )
) %>%
group_by(wave, model, horizon_wk) %>%
summarize(
  wis = mean(wis), mae=mean(abs_error),
  cov_50 = mean(coverage_50),
  cov_95 = mean(coverage_95)
)

```

'summarise()' has grouped output by 'wave', 'model'. You can override using the '.groups' argument.

```

wave_metrics_US <- wave_metrics_US %>%
mutate(
  rwis = case_when(
    horizon_wk == 1 ~ wis / pull(wave_metrics_US[9, 4]),
    horizon_wk == 2 ~ wis / pull(wave_metrics_US[10,4]),
    horizon_wk == 3 ~ wis / pull(wave_metrics_US[11,4]),
    horizon_wk == 4 ~ wis / pull(wave_metrics_US[12,4])
  ),
  rmae = case_when(
    horizon_wk == 1 ~ mae / pull(wave_metrics_US[9,5]),
    horizon_wk == 2 ~ mae / pull(wave_metrics_US[10,5]),
    horizon_wk == 3 ~ mae / pull(wave_metrics_US[11,5]),
    horizon_wk == 4 ~ mae / pull(wave_metrics_US[12,5])
  ),
  rcov_50 = case_when(
    horizon_wk == 1 ~ (cov_50-0.5) / abs(pull(wave_metrics_US[9,6])-0.5),
    horizon_wk == 2 ~ (cov_50-0.5) / abs(pull(wave_metrics_US[10,6])-0.5),
    horizon_wk == 3 ~ (cov_50-0.5) / abs(pull(wave_metrics_US[11,6])-0.5),
    horizon_wk == 4 ~ (cov_50-0.5) / abs(pull(wave_metrics_US[12,6])-0.5),
  ),
  rcov_95 = case_when(
    horizon_wk == 1 ~ (cov_95-0.95) / abs(pull(wave_metrics_US[9,7])-0.95),
    horizon_wk == 2 ~ (cov_95-0.95) / abs(pull(wave_metrics_US[10,7])-0.95),
    horizon_wk == 3 ~ (cov_95-0.95) / abs(pull(wave_metrics_US[11,7])-0.95),
    horizon_wk == 4 ~ (cov_95-0.95) / abs(pull(wave_metrics_US[12,7])-0.95),
  )
) %>%
mutate(across(where(is.numeric), round, digits=3))

knitr::kable(wave_metrics_US)

```

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	Base_arima_4root	1	298.271	478.368	0.100	0.833	0.803	1.477	-0.800	-2.333
alpha	Base_arima_4root	2	483.611	757.371	0.400	0.914	0.625	0.820	-0.206	-0.714
alpha	Base_arima_4root	3	647.750	1044.362	0.319	0.901	0.695	0.893	-0.371	-0.978
alpha	Base_arima_4root	4	827.597	1349.386	0.226	0.857	0.743	0.904	-0.575	-1.857
alpha	Base_arima_noTrans	1	364.629	648.507	0.100	1.000	0.981	2.002	-0.800	1.000
alpha	Base_arima_noTrans	2	563.022	863.548	0.514	1.000	0.727	0.935	0.029	1.000
alpha	Base_arima_noTrans	3	689.803	1088.364	0.549	1.000	0.740	0.931	0.101	1.000
alpha	Base_arima_noTrans	4	897.845	1508.112	0.440	1.000	0.807	1.011	-0.125	1.000
alpha	COVIDhub-baseline	1	371.549	323.933	1.000	1.000	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	2	774.310	923.143	0.986	1.000	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	3	932.443	1169.429	0.989	1.000	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	4	1113.129	1492.357	0.976	1.000	1.000	1.000	1.000	1.000
alpha	Multiple3_arima_4root	1	1400.673	1766.602	0.000	0.200	3.770	5.454	-1.000	-
										15.000
alpha	Multiple3_arima_4root	2	1224.000	1771.332	0.171	0.571	1.581	1.919	-0.676	-7.571
alpha	Multiple3_arima_4root	3	1391.170	2040.682	0.154	0.495	1.492	1.745	-0.708	-9.110
alpha	Multiple3_arima_4root	4	1782.333	2639.003	0.119	0.357	1.601	1.768	-0.800	-
										11.857
alpha	Multiple3_arima_noTrans	1	1620.340	2151.713	0.067	0.200	4.361	6.642	-0.867	-
										15.000
alpha	Multiple3_arima_noTrans	2	1472.230	2394.273	0.100	0.743	1.901	2.594	-0.824	-4.143
alpha	Multiple3_arima_noTrans	3	1692.610	2754.007	0.154	0.890	1.815	2.355	-0.708	-1.198
alpha	Multiple3_arima_noTrans	4	2187.866	3535.404	0.060	0.738	1.966	2.369	-0.925	-4.238
alpha	Topmost8_arima_4root	1	608.219	882.186	0.000	0.333	1.637	2.723	-1.000	-
										12.333
alpha	Topmost8_arima_4root	2	699.167	1062.503	0.286	0.686	0.903	1.151	-0.441	-5.286
alpha	Topmost8_arima_4root	3	887.183	1363.911	0.220	0.670	0.951	1.166	-0.573	-5.593
alpha	Topmost8_arima_4root	4	1072.725	1672.564	0.190	0.619	0.964	1.121	-0.650	-6.619
alpha	Topmost8_arima_noTrans	1	503.722	841.296	0.233	0.933	1.356	2.597	-0.533	-0.333
alpha	Topmost8_arima_noTrans	2	672.864	1170.013	0.457	1.000	0.869	1.267	-0.088	1.000
alpha	Topmost8_arima_noTrans	3	848.269	1468.658	0.484	1.000	0.910	1.256	-0.034	1.000
alpha	Topmost8_arima_noTrans	4	1011.064	1649.833	0.476	1.000	0.908	1.106	-0.050	1.000
winter21	Base_arima_4root	1	785.881	1195.135	0.133	0.767	2.115	3.689	-0.733	-3.667
winter21	Base_arima_4root	2	1211.615	1852.757	0.387	0.773	1.565	2.007	-0.233	-3.533
winter21	Base_arima_4root	3	1598.130	2389.736	0.352	0.762	1.714	2.044	-0.302	-3.762
winter21	Base_arima_4root	4	2159.655	3143.972	0.286	0.705	1.940	2.107	-0.450	-4.905
winter21	Base_arima_noTrans	1	1067.842	1478.342	0.067	0.567	2.874	4.564	-0.867	-7.667
winter21	Base_arima_noTrans	2	1326.824	1821.980	0.440	0.667	1.714	1.974	-0.124	-5.667
winter21	Base_arima_noTrans	3	1644.012	2264.901	0.438	0.714	1.763	1.937	-0.127	-4.714
winter21	Base_arima_noTrans	4	2074.327	2920.862	0.314	0.743	1.864	1.957	-0.390	-4.143
winter21	COVIDhub-baseline	1	836.762	1223.867	0.600	0.933	2.252	3.778	0.200	-0.333
winter21	COVIDhub-baseline	2	1300.857	2128.720	0.600	1.000	1.680	2.306	0.206	1.000
winter21	COVIDhub-baseline	3	1594.512	2589.971	0.552	1.000	1.710	2.215	0.107	1.000
winter21	COVIDhub-baseline	4	1929.342	3100.562	0.505	1.000	1.733	2.078	0.010	1.000

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	Multiple3_arima_4root	1	1894.871	2445.060	0.100	0.300	5.100	7.548	-0.800	-
										13.000
winter21	Multiple3_arima_4root	2	1557.360	2476.003	0.120	0.680	2.011	2.682	-0.782	-5.400
winter21	Multiple3_arima_4root	3	1438.387	2333.060	0.286	0.895	1.543	1.995	-0.438	-1.095
winter21	Multiple3_arima_4root	4	1748.996	2739.448	0.324	0.762	1.571	1.836	-0.370	-3.762
winter21	Multiple3_arima_noTrans	1	1825.491	2312.578	0.100	0.433	4.913	7.139	-0.800	-
										10.333
winter21	Multiple3_arima_noTrans	2	1391.443	2213.664	0.093	0.787	1.797	2.398	-0.837	-3.267
winter21	Multiple3_arima_noTrans	3	1299.539	2099.034	0.276	0.819	1.394	1.795	-0.458	-2.619
winter21	Multiple3_arima_noTrans	4	1681.499	2593.573	0.314	0.781	1.511	1.738	-0.390	-3.381
winter21	Topmost8_arima_4root	1	1623.849	2142.325	0.133	0.333	4.370	6.613	-0.733	-
										12.333
winter21	Topmost8_arima_4root	2	1366.081	2204.511	0.187	0.827	1.764	2.388	-0.645	-2.467
winter21	Topmost8_arima_4root	3	1493.815	2400.396	0.276	0.829	1.602	2.053	-0.458	-2.429
winter21	Topmost8_arima_4root	4	1789.474	2801.714	0.305	0.733	1.608	1.877	-0.410	-4.333
winter21	Topmost8_arima_noTrans	1	1609.830	2044.593	0.167	0.467	4.333	6.312	-0.667	-9.667
winter21	Topmost8_arima_noTrans	2	1554.891	2346.433	0.200	0.640	2.008	2.542	-0.618	-6.200
winter21	Topmost8_arima_noTrans	3	1688.000	2610.515	0.181	0.714	1.810	2.232	-0.652	-4.714
winter21	Topmost8_arima_noTrans	4	2050.529	3058.375	0.305	0.695	1.842	2.049	-0.410	-5.095

```
# Aggregated for all states and territories
wave_metrics_states <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location != "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    ),
    wave = case_when(
      forecast_date <= as.Date("2021-03-17") ~ "winter21",
      forecast_date >= as.Date("2021-03-17") ~ "alpha"
    )
  ) %>%
  group_by(wave, model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    cov_50 = mean(coverage_50),
    cov_95 = mean(coverage_95)
  )
```

'summarise()' has grouped output by 'wave', 'model'. You can override using the '.groups' argument.

```
wave_metrics_states <- wave_metrics_states %>%
  mutate(
    rwis = case_when(
      horizon_wk == 1 & wave == "alpha" ~ wis / pull(wave_metrics_states[9,4]),
```

```

horizon_wk == 2 & wave == "alpha" ~ wis / pull(wave_metrics_states[10,4]),
horizon_wk == 3 & wave == "alpha" ~ wis / pull(wave_metrics_states[11,4]),
horizon_wk == 4 & wave == "alpha" ~ wis / pull(wave_metrics_states[12,4]),
horizon_wk == 1 & wave == "winter21" ~ wis / pull(wave_metrics_states[37,4]),
horizon_wk == 2 & wave == "winter21" ~ wis / pull(wave_metrics_states[38,4]),
horizon_wk == 3 & wave == "winter21" ~ wis / pull(wave_metrics_states[39,4]),
horizon_wk == 4 & wave == "winter21" ~ wis / pull(wave_metrics_states[40,4])
),
rmae = case_when(
  horizon_wk == 1 & wave == "alpha" ~ mae / pull(wave_metrics_states[9,5]),
  horizon_wk == 2 & wave == "alpha" ~ mae / pull(wave_metrics_states[10,5]),
  horizon_wk == 3 & wave == "alpha" ~ mae / pull(wave_metrics_states[11,5]),
  horizon_wk == 4 & wave == "alpha" ~ mae / pull(wave_metrics_states[12,5]),
  horizon_wk == 1 & wave == "winter21" ~ mae / pull(wave_metrics_states[37,5]),
  horizon_wk == 2 & wave == "winter21" ~ mae / pull(wave_metrics_states[38,5]),
  horizon_wk == 3 & wave == "winter21" ~ mae / pull(wave_metrics_states[39,5]),
  horizon_wk == 4 & wave == "winter21" ~ mae / pull(wave_metrics_states[40,5])
),
rcov_50 = case_when(
  horizon_wk == 1 & wave == "alpha" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[9,6])-0.5),
  horizon_wk == 2 & wave == "alpha" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[10,6])-0.5),
  horizon_wk == 3 & wave == "alpha" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[11,6])-0.5),
  horizon_wk == 4 & wave == "alpha" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[12,6])-0.5),
  horizon_wk == 1 & wave == "winter21" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[37,6])-0.5),
  horizon_wk == 2 & wave == "winter21" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[38,6])-0.5),
  horizon_wk == 3 & wave == "winter21" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[39,6])-0.5),
  horizon_wk == 4 & wave == "winter21" ~ (cov_50-0.5) / abs(pull(wave_metrics_states[40,6])-0.5)
),
rcov_95 = case_when(
  horizon_wk == 1 & wave == "alpha" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[9,7])-0.95),
  horizon_wk == 2 & wave == "alpha" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[10,7])-0.95),
  horizon_wk == 3 & wave == "alpha" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[11,7])-0.95),
  horizon_wk == 4 & wave == "alpha" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[12,7])-0.95),
  horizon_wk == 1 & wave == "winter21" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[37,7])-0.95),
  horizon_wk == 2 & wave == "winter21" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[38,7])-0.95),
  horizon_wk == 3 & wave == "winter21" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[39,7])-0.95),
  horizon_wk == 4 & wave == "winter21" ~ (cov_95-0.95) / abs(pull(wave_metrics_states[40,7])-0.95)
)
) %>%
mutate(across(where(is.numeric), round, digits=3))

knitr::kable(wave_metrics_states)

```

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	Base_arima_4root	1	8.306	12.731	0.512	0.906	0.624	0.961	0.026	-1.000
alpha	Base_arima_4root	2	12.789	19.860	0.504	0.946	0.548	0.806	0.010	-0.080
alpha	Base_arima_4root	3	16.394	25.759	0.502	0.955	0.600	0.877	0.005	0.109
alpha	Base_arima_4root	4	20.933	32.675	0.490	0.950	0.662	0.917	-0.023	0.002
alpha	Base_arima_noTrans	1	10.440	16.125	0.651	0.987	0.784	1.217	0.337	0.838
alpha	Base_arima_noTrans	2	15.818	24.530	0.706	0.993	0.678	0.995	0.447	0.969
alpha	Base_arima_noTrans	3	19.082	29.607	0.728	0.990	0.699	1.009	0.494	0.918
alpha	Base_arima_noTrans	4	23.545	36.957	0.715	0.989	0.745	1.037	0.472	0.877

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
alpha	COVIDhub-baseline	1	13.308	13.254	0.947	0.994	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	2	23.331	24.641	0.961	0.995	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	3	27.315	29.355	0.961	0.994	1.000	1.000	1.000	1.000
alpha	COVIDhub-baseline	4	31.616	35.644	0.956	0.995	1.000	1.000	1.000	1.000
alpha	Multiple3_arima_4root	1	19.934	28.371	0.315	0.712	1.498	2.141	-0.415	-5.471
alpha	Multiple3_arima_4root	2	26.861	35.382	0.346	0.840	1.151	1.436	-0.334	-2.469
alpha	Multiple3_arima_4root	3	49.094	40.979	0.361	0.868	1.797	1.396	-0.302	-1.870
alpha	Multiple3_arima_4root	4	145.358	53.066	0.333	0.857	4.598	1.489	-0.367	-2.097
alpha	Multiple3_arima_noTrans	1	26.375	39.816	0.281	0.789	1.982	3.004	-0.489	-3.691
alpha	Multiple3_arima_noTrans	2	28.797	49.045	0.328	0.968	1.234	1.990	-0.374	0.395
alpha	Multiple3_arima_noTrans	3	32.778	56.299	0.356	0.977	1.200	1.918	-0.312	0.622
alpha	Multiple3_arima_noTrans	4	41.065	70.591	0.333	0.977	1.299	1.980	-0.366	0.609
alpha	Topmost8_arima_4root	1	12.882	19.407	0.388	0.799	0.968	1.464	-0.251	-3.456
alpha	Topmost8_arima_4root	2	15.551	24.841	0.411	0.907	0.667	1.008	-0.194	-0.969
alpha	Topmost8_arima_4root	3	19.134	30.761	0.411	0.929	0.700	1.048	-0.193	-0.491
alpha	Topmost8_arima_4root	4	23.110	37.052	0.407	0.936	0.731	1.039	-0.204	-0.317
alpha	Topmost8_arima_noTrans	1	12.375	19.697	0.563	0.966	0.930	1.486	0.141	0.368
alpha	Topmost8_arima_noTrans	2	16.369	24.147	0.702	0.989	0.702	0.980	0.439	0.870
alpha	Topmost8_arima_noTrans	3	19.691	29.793	0.704	0.992	0.721	1.015	0.442	0.956
alpha	Topmost8_arima_noTrans	4	23.027	35.651	0.716	0.996	0.728	1.000	0.475	1.026
winter21	Base_arima_4root	1	27.338	38.404	0.496	0.877	1.144	1.158	-0.020	-9.500
winter21	Base_arima_4root	2	34.898	53.760	0.469	0.911	0.947	1.024	-0.092	-0.917
winter21	Base_arima_4root	3	43.765	66.848	0.460	0.896	0.988	1.044	-0.117	-1.278
winter21	Base_arima_4root	4	54.503	82.352	0.441	0.871	1.045	1.072	-0.175	-1.986
winter21	Base_arima_noTrans	1	30.227	41.441	0.395	0.819	1.265	1.250	-0.462	-
										17.083
winter21	Base_arima_noTrans	2	35.651	53.217	0.457	0.866	0.967	1.013	-0.129	-1.940
winter21	Base_arima_noTrans	3	44.023	64.730	0.470	0.856	0.994	1.011	-0.088	-2.230
winter21	Base_arima_noTrans	4	53.433	77.649	0.474	0.839	1.025	1.011	-0.077	-2.784
winter21	COVIDhub-baseline	1	23.898	33.164	0.728	0.958	1.000	1.000	1.000	1.000
winter21	COVIDhub-baseline	2	36.858	52.513	0.835	0.993	1.000	1.000	1.000	1.000
winter21	COVIDhub-baseline	3	44.295	64.016	0.839	0.992	1.000	1.000	1.000	1.000
winter21	COVIDhub-baseline	4	52.140	76.788	0.839	0.990	1.000	1.000	1.000	1.000
winter21	Multiple3_arima_4root	1	51.049	65.913	0.320	0.728	2.136	1.987	-0.792	-
										28.833
winter21	Multiple3_arima_4root	2	41.558	62.218	0.374	0.864	1.128	1.185	-0.377	-1.994
winter21	Multiple3_arima_4root	3	41.464	63.327	0.420	0.898	0.936	0.989	-0.236	-1.230
winter21	Multiple3_arima_4root	4	47.184	72.727	0.418	0.887	0.905	0.947	-0.241	-1.583
winter21	Multiple3_arima_noTrans	1	43.933	55.868	0.349	0.734	1.838	1.685	-0.662	-
										28.083
winter21	Multiple3_arima_noTrans	2	37.741	56.352	0.413	0.893	1.024	1.073	-0.259	-1.333
winter21	Multiple3_arima_noTrans	3	39.663	60.769	0.454	0.908	0.895	0.949	-0.136	-1.009
winter21	Multiple3_arima_noTrans	4	47.682	73.824	0.442	0.895	0.914	0.961	-0.170	-1.385

wave	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae	rcov_50	rcov_95
winter21	Topmost8_arima_4root	1	41.044	54.985	0.377	0.778	1.717	1.658	-0.541	-
										22.417
winter21	Topmost8_arima_4root	2	36.797	56.152	0.418	0.897	0.998	1.069	-0.245	-1.226
winter21	Topmost8_arima_4root	3	40.706	62.737	0.436	0.902	0.919	0.980	-0.188	-1.135
winter21	Topmost8_arima_4root	4	47.791	73.280	0.440	0.885	0.917	0.954	-0.177	-1.619
winter21	Topmost8_arima_noTrans	1	39.551	51.127	0.365	0.758	1.655	1.542	-0.594	-
										25.000
winter21	Topmost8_arima_noTrans	2	38.427	58.162	0.372	0.869	1.043	1.108	-0.384	-1.875
winter21	Topmost8_arima_noTrans	3	43.334	66.696	0.384	0.872	0.978	1.042	-0.342	-1.848
winter21	Topmost8_arima_noTrans	4	49.975	76.428	0.414	0.858	0.958	0.995	-0.254	-2.317

For US National during the alpha wave, the Base_arima_noTrans model performed the best for coverage rates and second best for WIS and MAE at every horizon. The Base_arima_4root model performed best in terms of WIS and MAE but was third for coverage rate while the Topmost8_arima_noTrans model had the second best coverage rates and third best WIS and MAE. During the alpha wave, these models performed the best and tended to have more consistent values for their metrics. The winter21 wave saw noticeably worse values for all metrics at every horizon. Generally, the two models that use the base set hierarchy tended to perform the best for all metrics, although the coverage rates varied a lot by horizon, with the Topmost8_arima_4root and Multiple3_arima_noTrans achieving closer to nominal coverage rates for the 95% interval than even the Base models at horizons greater than 1 week ahead. These results are pretty consistent with improvements over the base-line, which ranged from modest to significant for the best performing models during the alpha wave, especially for WIS and 50% PI coverage. The winter21 wave saw...

For states during both waves, we again see noticeably better coverage rates that approach the nominal level (or even exceed it) for all of the models, compared to those of the US National level when broken into waves. During the alpha wave, the Base_arima_4root model has the best performance for all metrics across all horizons. It is followed closely by the Topmost8_arima_noTrans and Base_arima_noTrans models, then the Topmost8_arima_4root and Multiple3_arima_noTrans models, all of which are much better than the Multiple3_arima_4root model. During the winter21 wave, the Base_arima_4root model is again generally the best for all metrics at all horizon weeks but sometimes loses out in terms of 95% coverage at longer horizons to other models. Notably during this wave, there are much greater similarities in coverage rates and certain models have better coverage rates for the 50% interval while others have better coverage reads for the 95% one.

By forecast date (week)

```
# US National
forecast_date_metrics_US <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location == "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    )
  )
```

```

) %>%
group_by(forecast_date, model, horizon_wk) %>%
summarize(
  wis = mean(wis), mae=mean(abs_error),
  cov_50 = mean(coverage_50),
  cov_95 = mean(coverage_95)
)

```

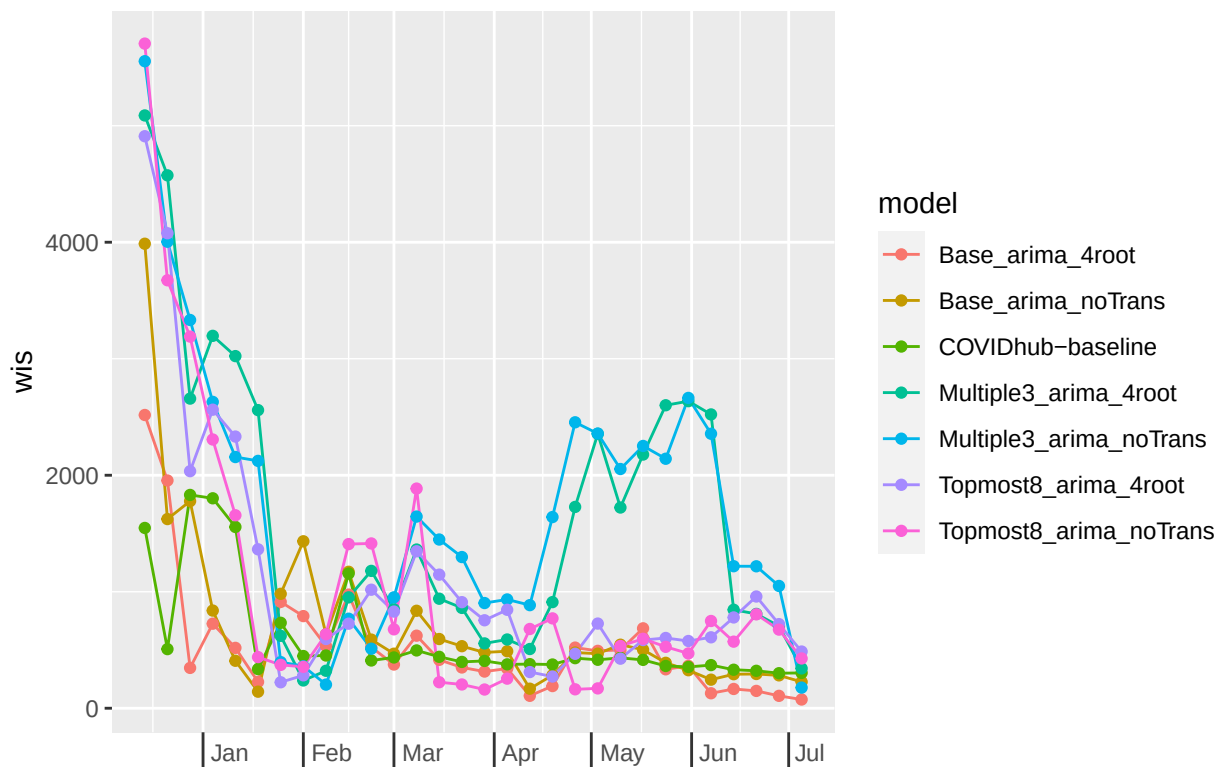
'summarise()' has grouped output by 'forecast_date', 'model'. You can override using the '.groups' a

```

# WIS
ggplot(filter(forecast_date_metrics_US, horizon_wk == 1),
  aes(x = forecast_date + weeks(1), y = wis)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 1-week ahead WIS by model (US)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
  axis.text.x = element_text(vjust = 7, hjust = -0.2))

```

Average 1-week ahead WIS by model (US)

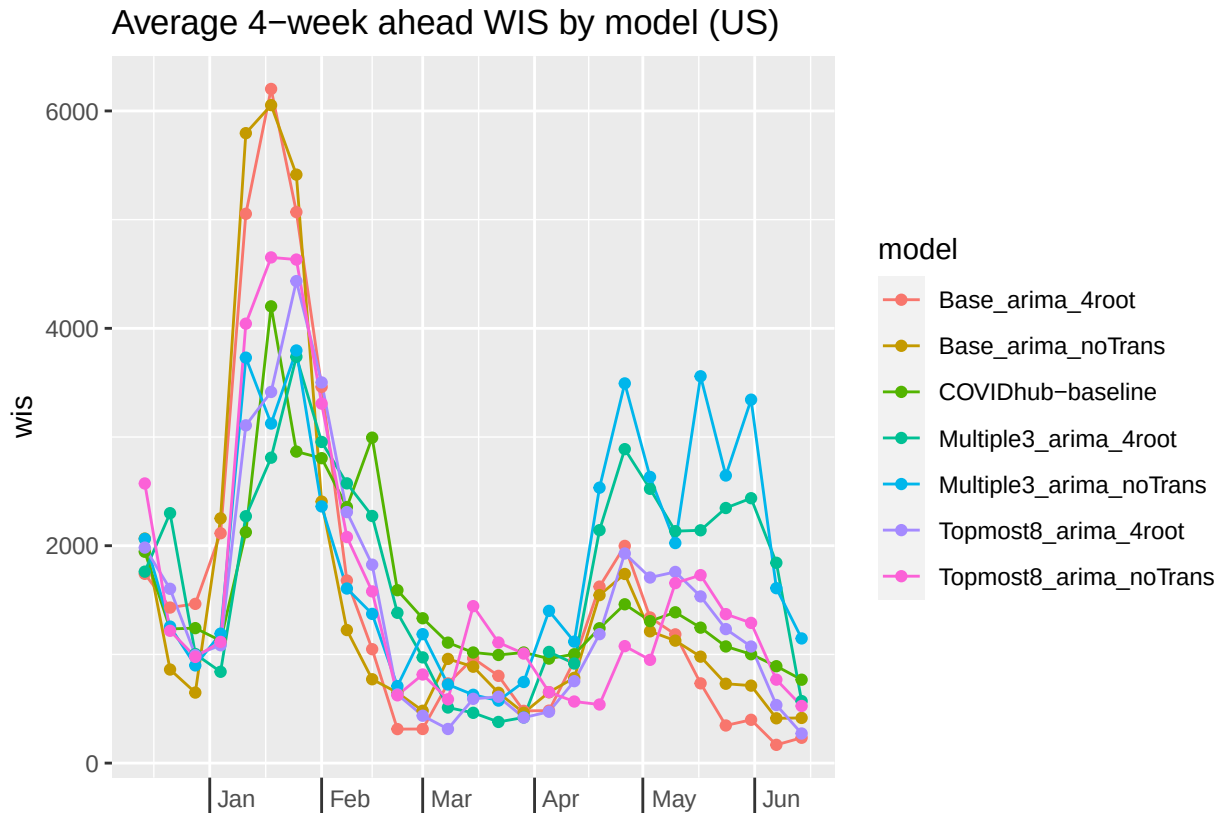


```

ggplot(filter(forecast_date_metrics_US, horizon_wk == 4),
  aes(x = forecast_date + weeks(1), y = wis)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +

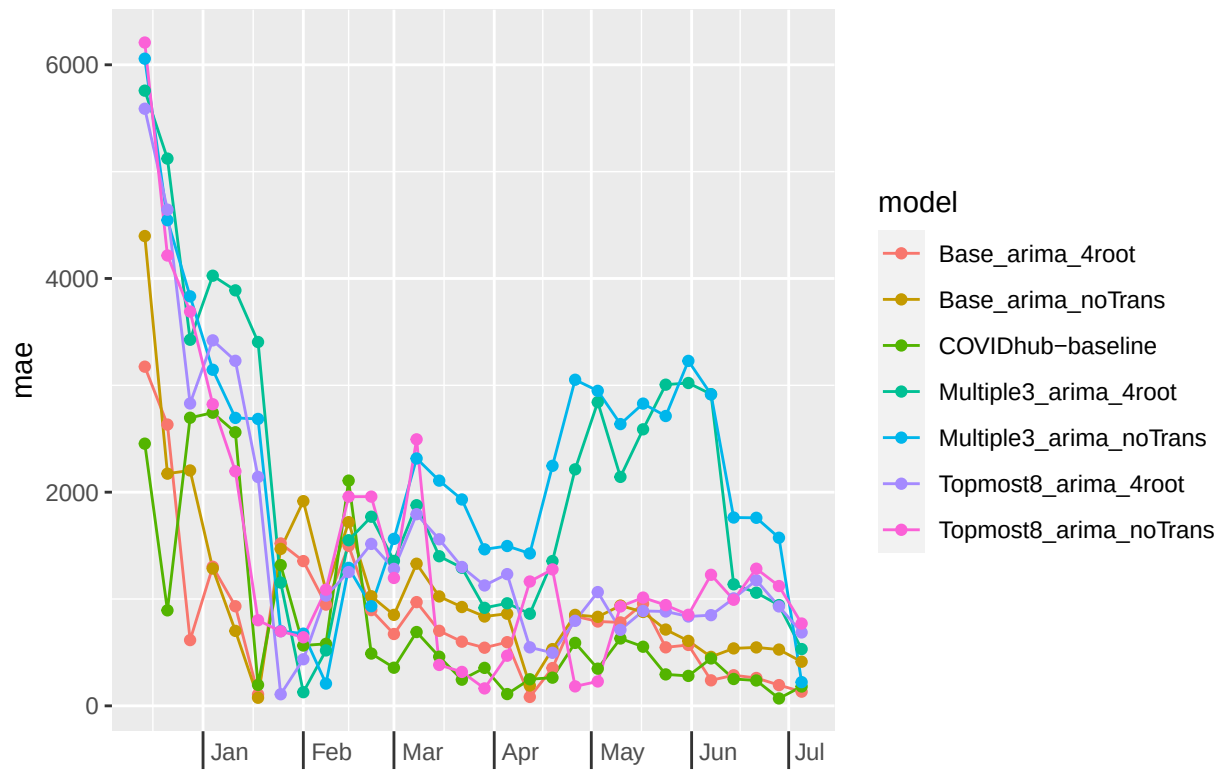
```

```
scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
labs(title = "Average 4-week ahead WIS by model (US)") +
theme(axis.ticks.length.x = unit(0.5, "cm"),
      axis.text.x = element_text(vjust = 7, hjust = -0.2))
```



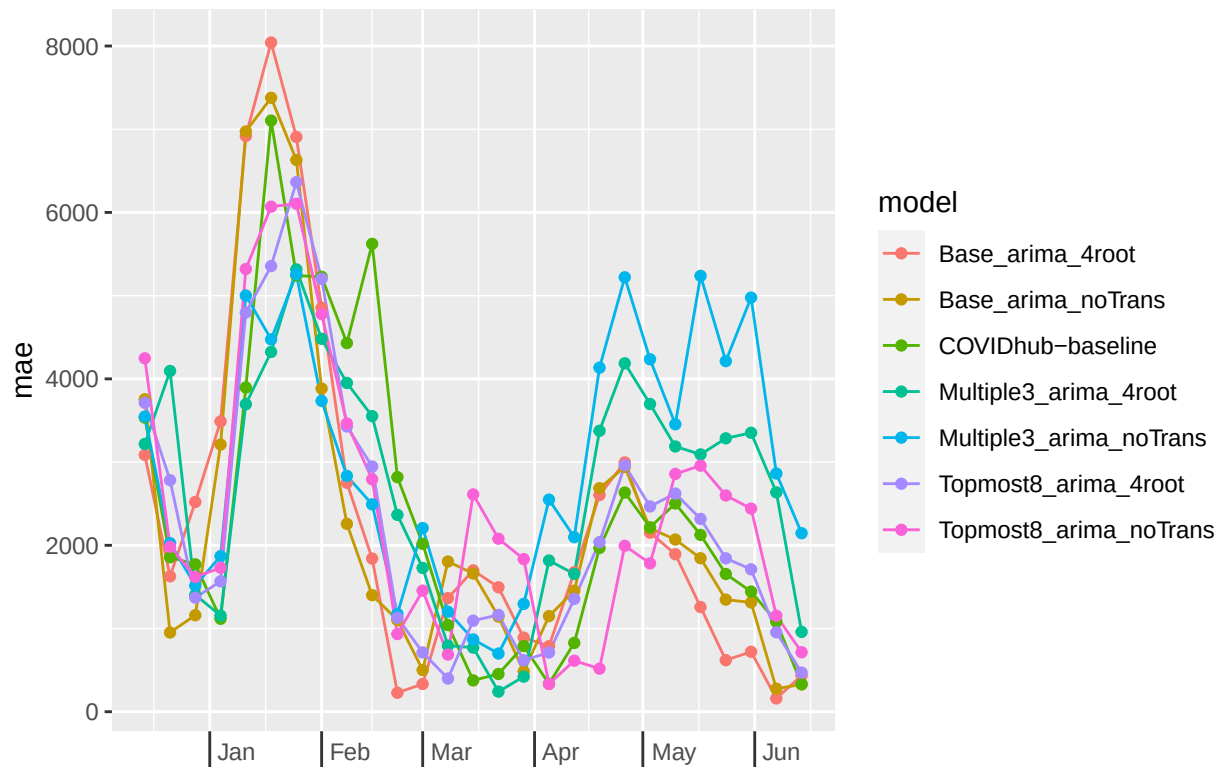
```
# MAE
ggplot(filter(forecast_date_metrics_US, horizon_wk == 1),
      aes(x = forecast_date + weeks(1), y = mae)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 1-week ahead MAE by model (US)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
        axis.text.x = element_text(vjust = 7, hjust = -0.2))
```


Average 1-week ahead MAE by model (US)

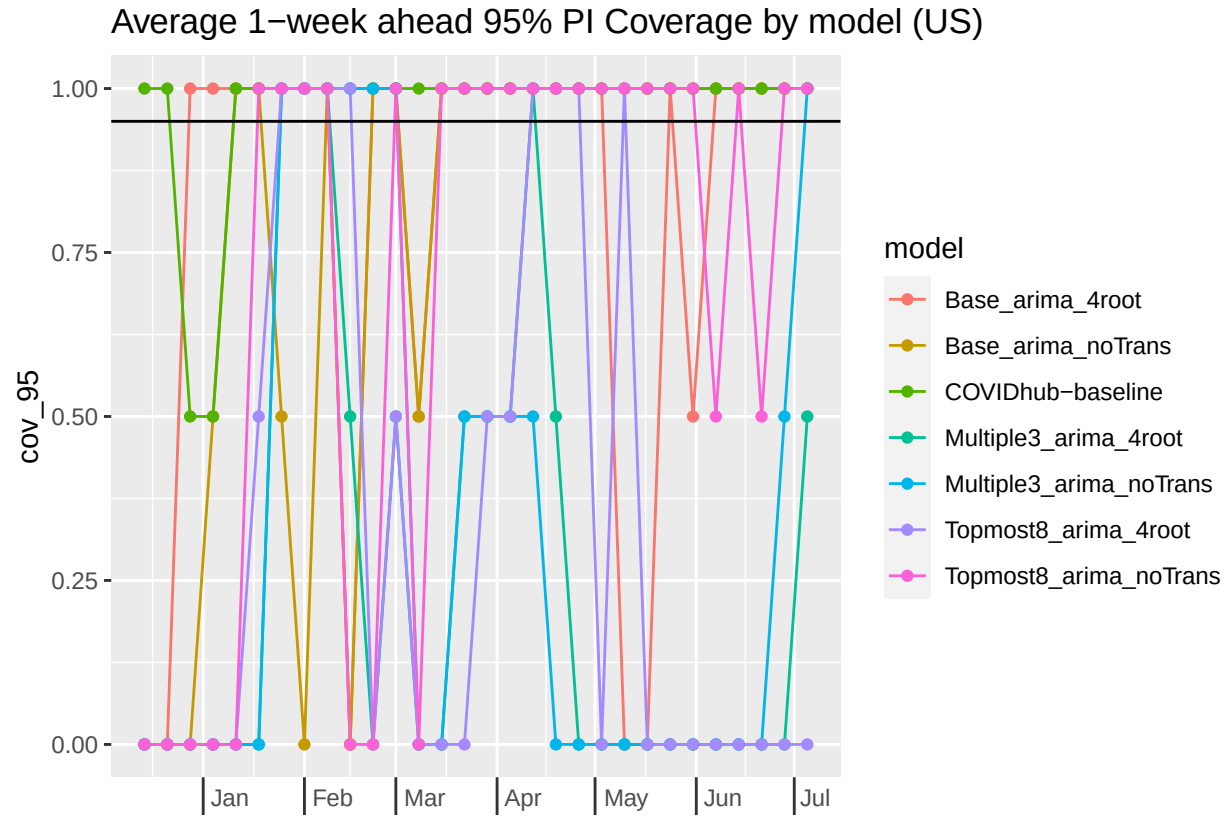


```
ggplot(filter(forecast_date_metrics_US, horizon_wk == 4),
  aes(x = forecast_date + weeks(1), y = mae)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 4-week ahead MAE by model (US)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```

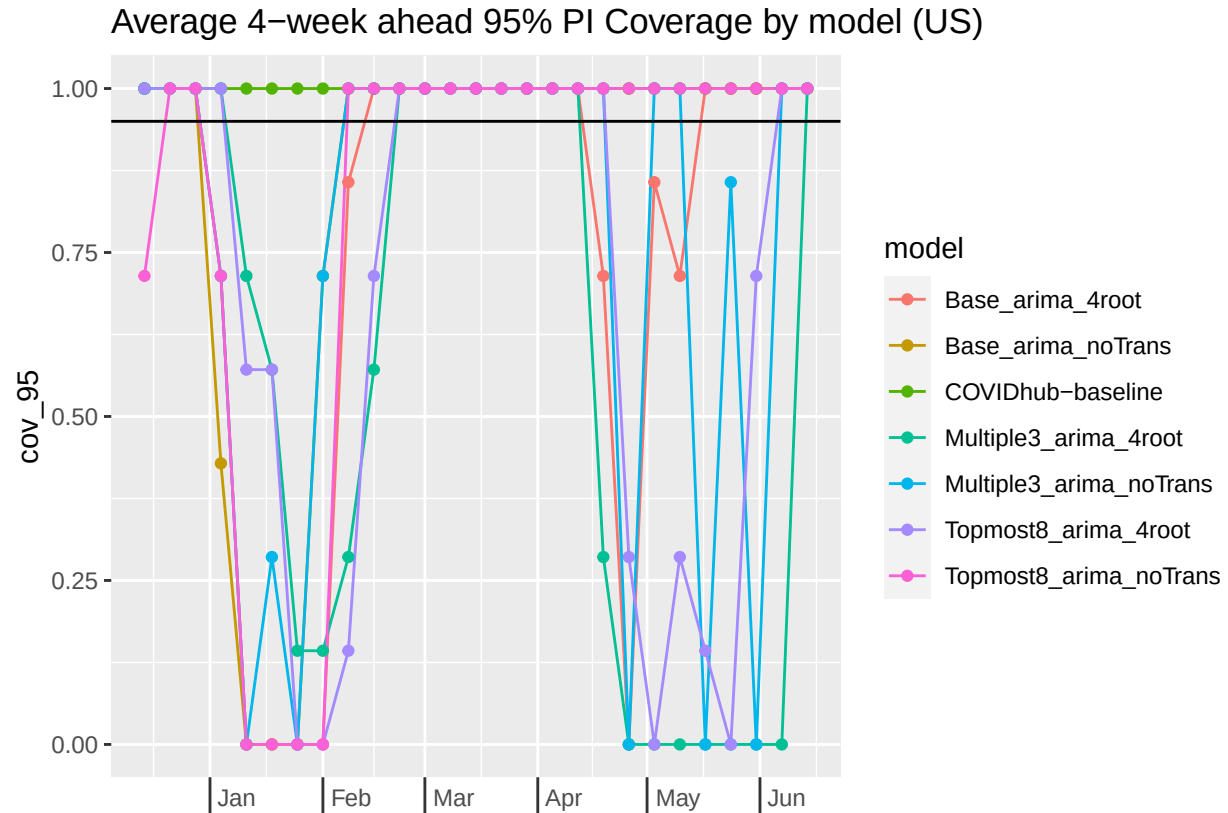
Average 4-week ahead MAE by model (US)



```
# 95% PI Coverage
ggplot(filter(forecast_date_metrics_US, horizon_wk == 1),
  aes(x = forecast_date + weeks(1), y = cov_95)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  geom_hline(aes(yintercept=0.95)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 1-week ahead 95% PI Coverage by model (US)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```



```
ggplot(filter(forecast_date_metrics_US, horizon_wk == 4),
  aes(x = forecast_date + weeks(1), y = cov_95)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  geom_hline(aes(yintercept=0.95)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 4-week ahead 95% PI Coverage by model (US)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```

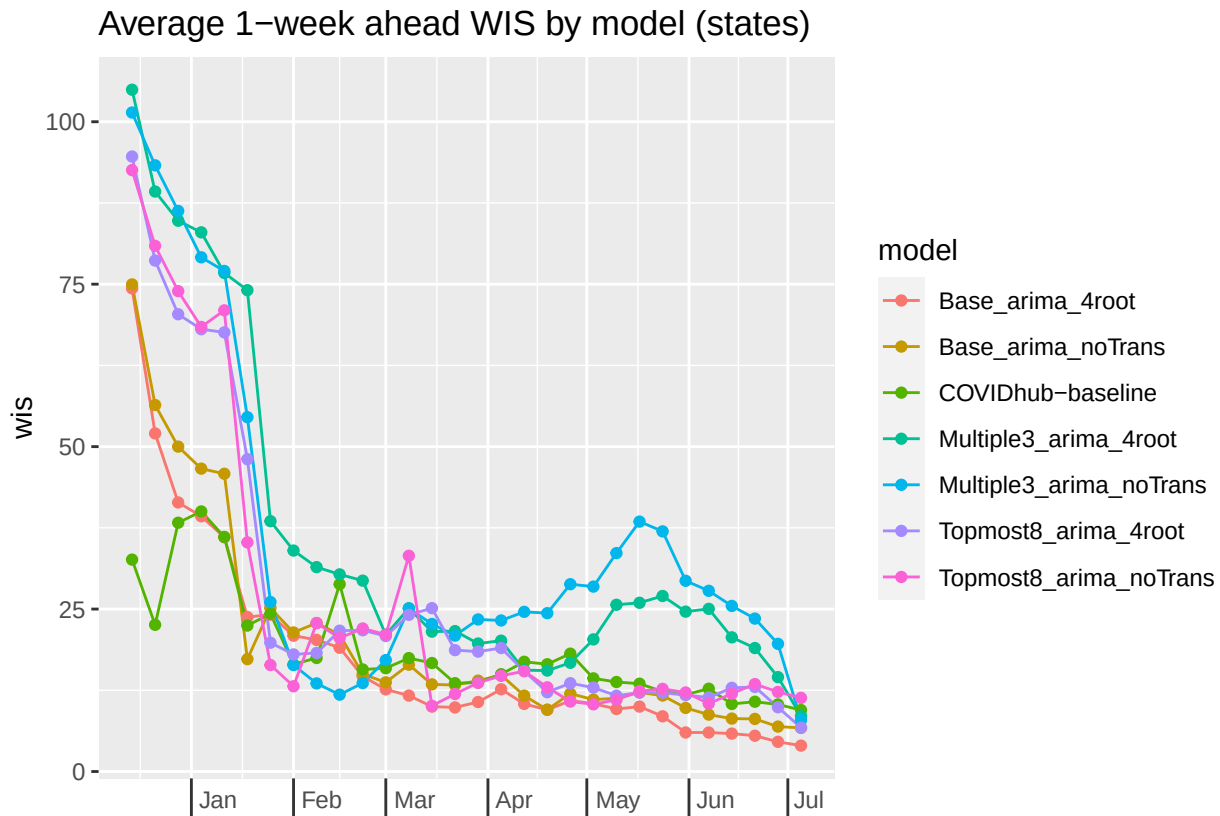


```
# Aggregated for all states and territories
forecast_date_metrics_states <- scores_df %>%
  rbind(score_baseline) %>%
  filter(horizon <= 28,
         target_end_date <= as.Date("2021-07-05"),
         location != "US") %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    ) %>%
  group_by(forecast_date, model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    cov_50 = mean(coverage_50),
    cov_95 = mean(coverage_95)
  )
```

'summarise()' has grouped output by 'forecast_date', 'model'. You can override using the '.groups' a

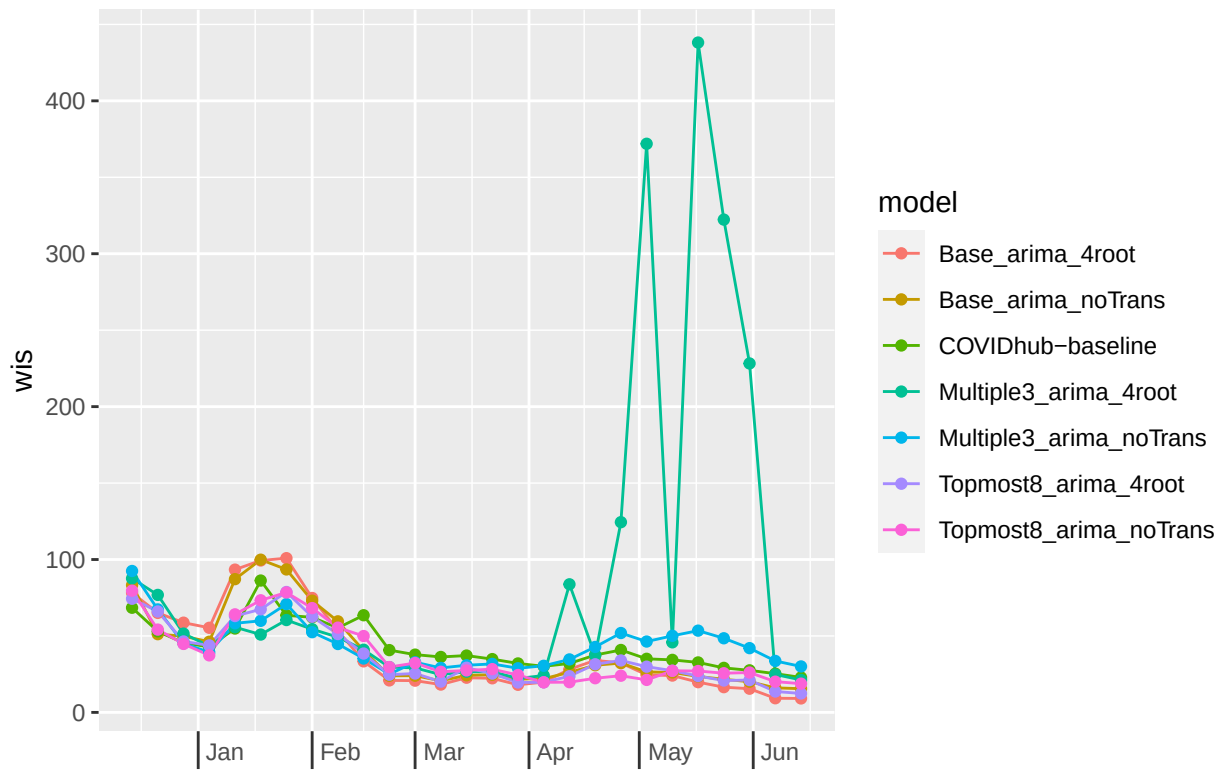
```
# WIS
ggplot(filter(forecast_date_metrics_states, horizon_wk == 1),
       aes(x = forecast_date + weeks(1), y = wis)) +
```

```
geom_line(aes(color=model)) +
geom_point(aes(color=model)) +
scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
labs(title = "Average 1-week ahead WIS by model (states)") +
theme(axis.ticks.length.x = unit(0.5, "cm"),
      axis.text.x = element_text(vjust = 7, hjust = -0.2))
```



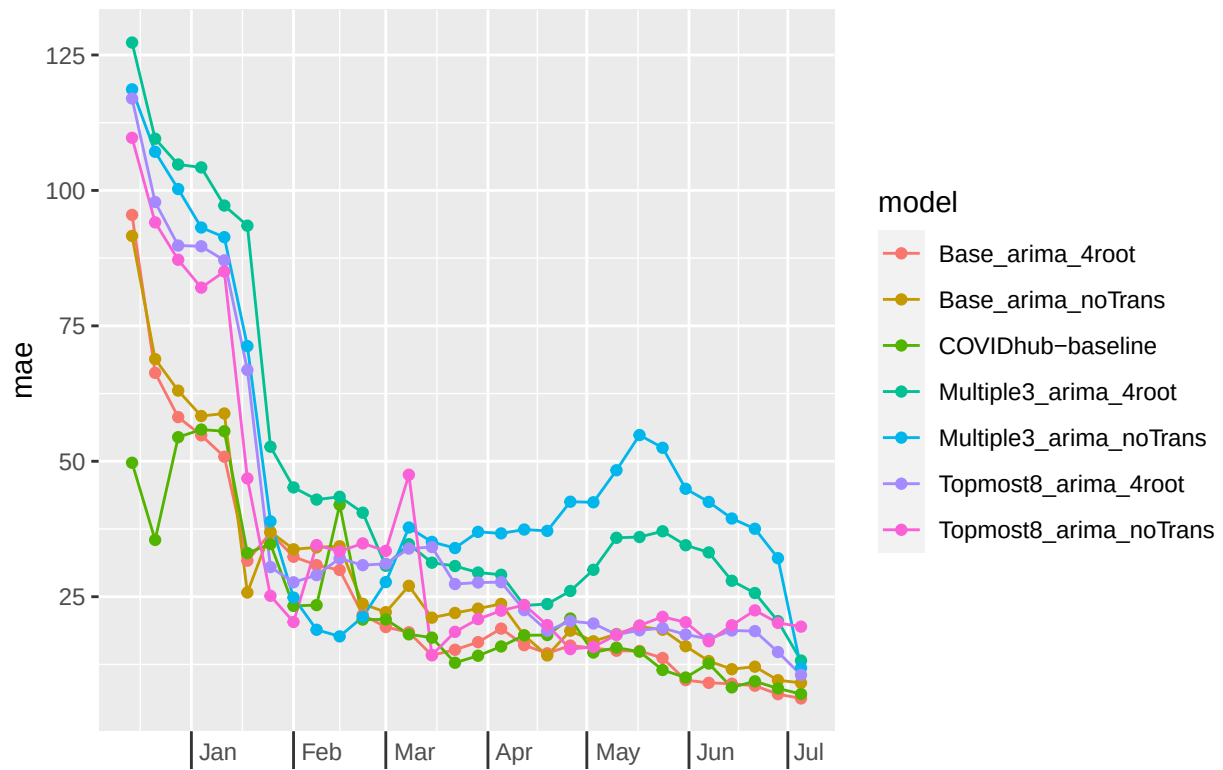
```
ggplot(filter(forecast_date_metrics_states, horizon_wk == 4),
      aes(x = forecast_date + weeks(1), y = wis)) +
geom_line(aes(color=model)) +
geom_point(aes(color=model)) +
scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
labs(title = "Average 4-week ahead WIS by model (states)") +
theme(axis.ticks.length.x = unit(0.5, "cm"),
      axis.text.x = element_text(vjust = 7, hjust = -0.2))
```

Average 4-week ahead WIS by model (states)



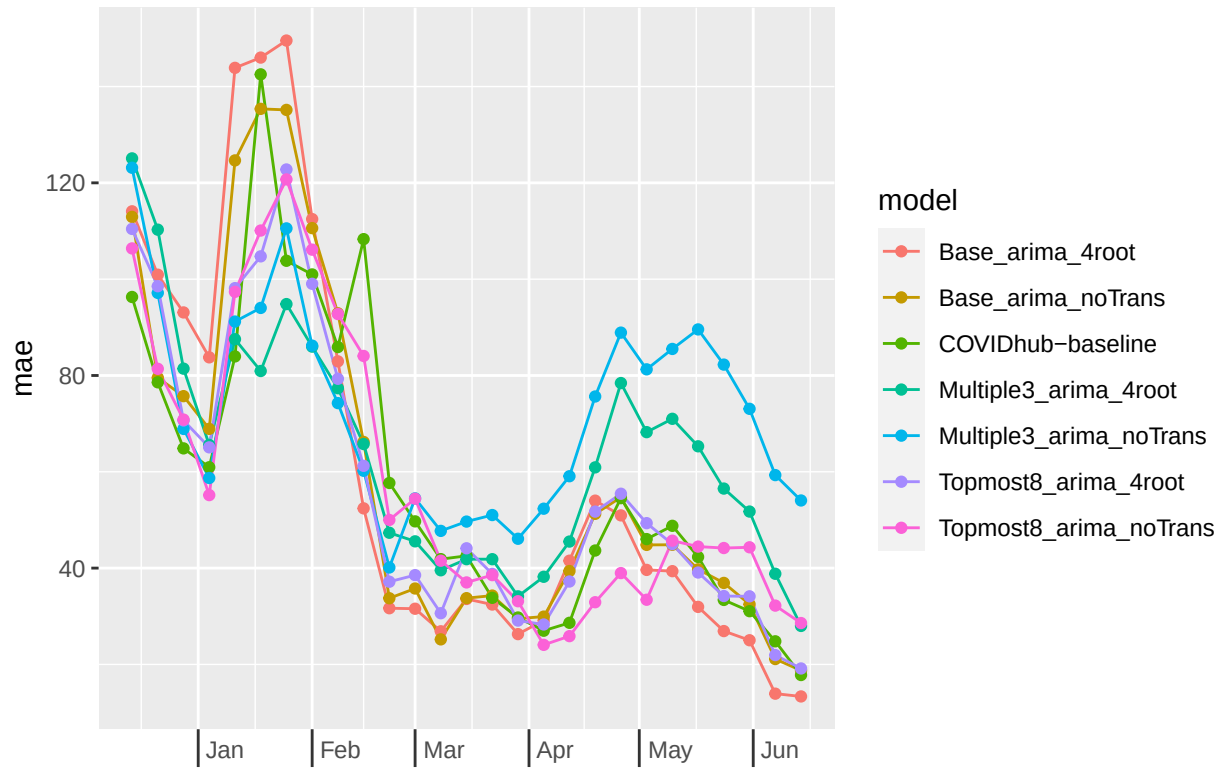
```
# MAE
ggplot(filter(forecast_date_metrics_states, horizon_wk == 1),
  aes(x = forecast_date + weeks(1), y = mae)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 1-week ahead MAE by model (states)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```

Average 1-week ahead MAE by model (states)



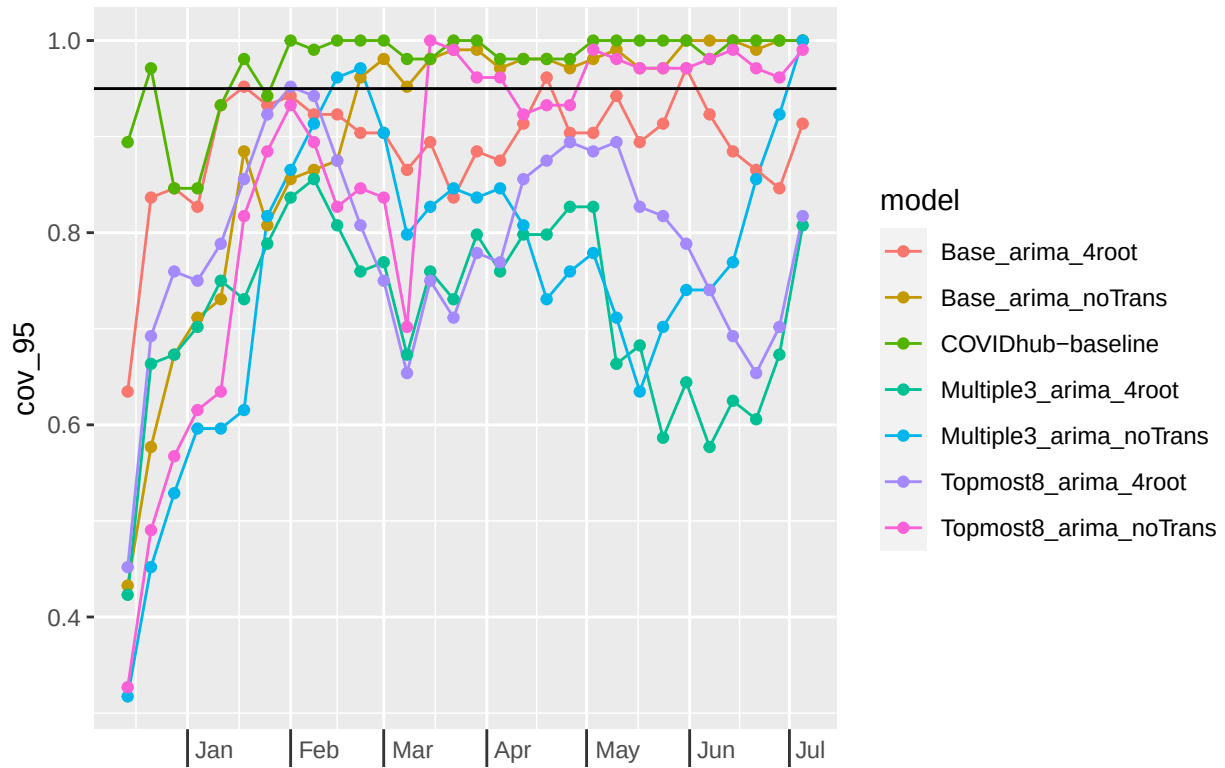
```
ggplot(filter(forecast_date_metrics_states, horizon_wk == 4),
  aes(x = forecast_date + weeks(1), y = mae)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 4-week ahead MAE by model (states)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```

Average 4-week ahead MAE by model (states)

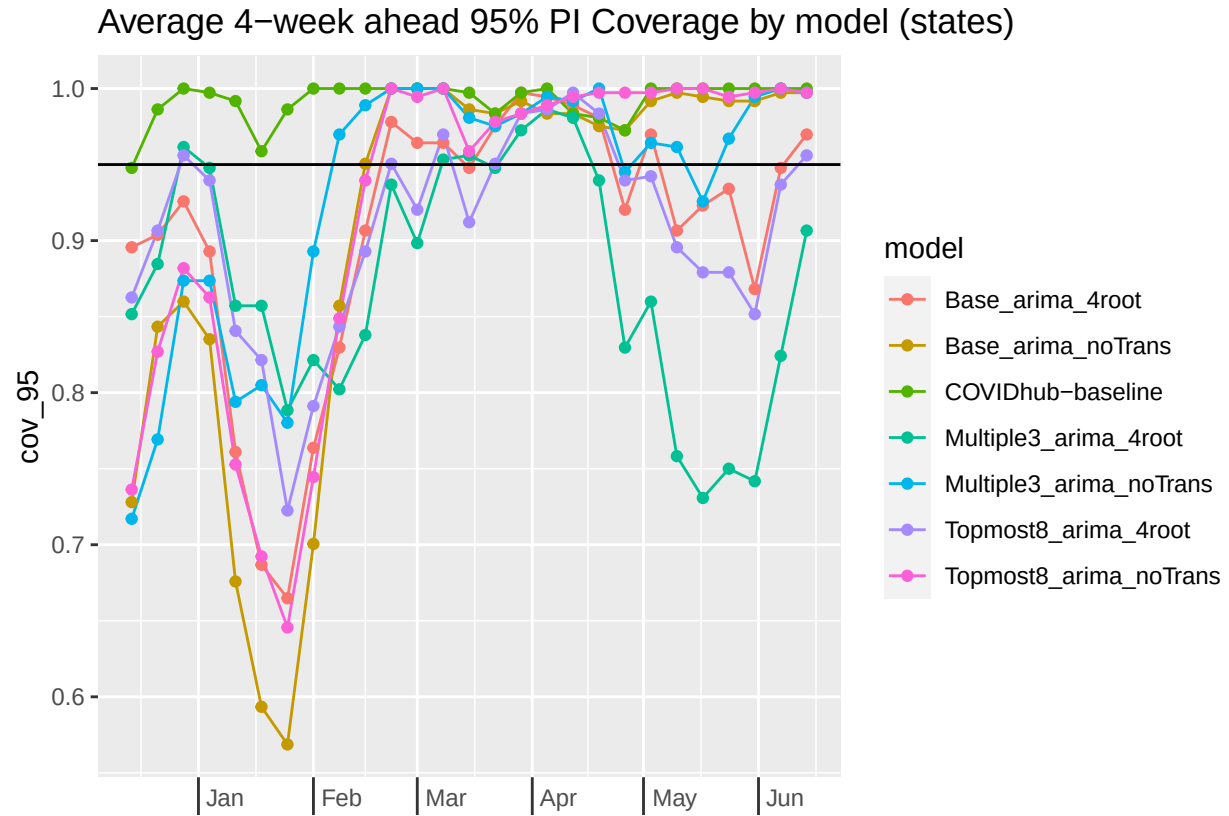


```
# 95% PI Coverage
ggplot(filter(forecast_date_metrics_states, horizon_wk == 1),
  aes(x = forecast_date + weeks(1), y = cov_95)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  geom_hline(aes(yintercept=0.95)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 1-week ahead 95% PI Coverage by model (states)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```


Average 1-week ahead 95% PI Coverage by model (states)



```
ggplot(filter(forecast_date_metrics_states, horizon_wk == 4),
  aes(x = forecast_date + weeks(1), y = cov_95)) +
  geom_line(aes(color=model)) +
  geom_point(aes(color=model)) +
  geom_hline(aes(yintercept=0.95)) +
  scale_x_date(name=NULL, date_breaks = "1 month", date_labels = "%b") +
  labs(title = "Average 4-week ahead 95% PI Coverage by model (states)") +
  theme(axis.ticks.length.x = unit(0.5, "cm"),
    axis.text.x = element_text(vjust = 7, hjust = -0.2))
```



Scoring Conclusions

It seems that the best three models that should be passed onto the testing phase are the Base_arma_noTrans, the Base_arma_4root, and the Topmost8_arma_noTrans models, which seem to take turns being most accurate depending on state or national level overall and during different waves.